

Decodable but Not Accessible: Auditing Distance-Based Reliability Estimation on Disentangled Skin-Lesion Representations

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Abstract

Distance-based reliability estimation assumes that a representation’s geometry reflects its trustworthiness, yet this assumption is rarely tested under training interventions that reshape representation geometry directly. We audit this assumption under domain-adversarial representation learning, using a disentanglement dose-response ladder. Three checkpoint families share the same architecture and a 16-dimensional representation, differing only in orthogonality strength ($\lambda_{orth} = 0, 1, 5$). Representation geometry changed substantially with disentanglement strength: condition number shifted by two orders of magnitude (Kendall’s $\tau = 0.84$, exact $p = 2.8 \times 10^{-5}$). This geometric change was not accompanied by any improvement in reliability estimation: Mahalanobis-distance AUROC (ISIC-test vs. PAD-UFES) remained flat and below chance (≈ 0.40) at every disentanglement level, with no significant association to any of five geometry metrics tested. The same failure was shared by cosine-to-centroid and pooled k -nearest-neighbor scorers, as well as three additional, non-distance-based scorers (an energy-based confidence score, Virtual-Logit Matching, and a kernel density estimator) – eight scorers spanning distinct scoring principles, seven of which converged on the same result; one (the energy-based score) showed an isolated upward trend with disentanglement strength that we report but do not treat as evidence against the overall pattern. A supervised probe with no access to the training objective nonetheless recovered domain membership from the identical embeddings at 0.72–0.81 AUROC across every disentanglement level, showing the relevant information was not absent from the representation. Collectively, these findings indicate that evaluating a learned representation by classification performance alone can overlook whether the information it contains is organized in a form downstream reliability estimators can actually use. Information can remain decodable while becoming largely inaccessible to non-probing reliability estimators.

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1. Introduction

Deep neural networks are typically evaluated by predictive accuracy, yet deployment safety also depends on the structure of the representation a network learns internally, not only on how it classifies a fixed test set. Whether downstream reliability estimation succeeds therefore depends not only on what a representation encodes, but also on how that information is organized. A representation’s geometry—how tightly points cluster by class, how well-conditioned the resulting covariance is, how closely that structure resembles the class-conditional Gaussian shape many downstream estimators assume—is itself shaped by training, and can change substantially without moving classification accuracy at all.

Distance-based reliability and out-of-distribution (OOD) estimation methods are built directly on this geometry. Mahalanobis distance [?] scores a test point by its distance to the nearest class-conditional mean under an estimated covariance; cosine-similarity-to-centroid variants replace the covariance-weighted distance with a simpler angular one; non-parametric k -nearest-neighbor scorers [?] drop the notion of a class centroid altogether and use local density in the representation space instead. Despite spanning fully parametric to fully non-parametric, all three rest on the same premise: whatever geometric structure a representation has, a sufficiently faithful measure of distance within it will track how typical, and therefore how trustworthy, a given input is. Other reliability estimators relax this purely geometric premise—Energy [?] and Virtual-Logit Matching [?] incorporate the classifier’s own logits directly, and density-based estimators replace a single Gaussian assumption with a non-parametric one—but share the same underlying constraint: all are computed from a fixed, already-trained representation, without retraining or access to the original objective.

This premise ties reliability estimation to representation geometry by construction, so any training procedure that reshapes representation geometry also reshapes, whether or not this is intended, the reliability estimator built on top of it. Domain-adversarial and orthogonality-based disentanglement training [?] are procedures designed explicitly to reorganize this geometry—separating or suppressing structure associated with a nuisance variable such as acquisition site or imaging domain—and are increasingly used in clinical imaging pipelines for exactly that reason. Whether the geometric changes such training induces are the kind distance-based reliability estimation can still read, or the kind that severs the link between geometry and reliability without affecting classification accuracy, has not, to our knowledge, been audited directly.

We audit this assumption using a disentanglement dose-response ladder that isolates the effect of orthogonality strength while holding the architecture and representation fixed. This design allows us to ask whether geometric change,

distance-based reliability estimation, and information decodability remain coupled under disentanglement. Doing so lets us distinguish two claims the existing OOD literature does not, to our knowledge, separate for a training-induced geometric shift: that a representation has lost the information a reliability estimator would need, versus that the representation still holds this information in a form distance-based geometry cannot use.

Across the ladder, representation geometry changes substantially, whereas eight structurally distinct reliability estimators—five distance-based, three not—consistently remain near or below chance (~ 0.40 AUROC), with a single exception discussed as a hypothesis-generating observation rather than a counter-finding (Section 4). A supervised probe applied to the identical representations, in contrast, recovers the relevant domain signal at 0.73–0.81 AUROC. These results are obtained within a domain-adversarial training regime evaluated against its own adversarial target domain (Section 2.2), and are scoped accordingly throughout this paper. Information can remain decodable while becoming largely inaccessible to non-probing reliability estimators.

2. Methods

2.1. Overall study design

For each of 13 checkpoints (Section 2.3), we extract z_{lesion} embeddings for the ISIC training set and for the ISIC-test/PAD-UFES evaluation split. From these embeddings we compute three geometry metrics on the fitted Mahalanobis parameters (Section 2.4), score the evaluation split with eight reliability estimators built on the identical embeddings—five distance-based, three not (Section 2.5)—and train three supervised probes on the same embeddings to quantify domain-information decodability independent of any of these scorers (Section 2.6). Association between disentanglement strength, geometry, reliability estimation, and decodability is then tested across the ladder using exact-permutation statistics (Section 2.7). Figure 1 summarizes this pipeline.

2.2. Datasets and domains

We use two publicly available skin lesion datasets: ISIC 2018 [? ?] as the source domain and PAD-UFES-20 [?] as the target domain, following the split and preprocessing pipeline of the underlying disentanglement classifier this study audits. ISIC data is partitioned into training, validation, and test subsets under a fixed, seeded split. PAD-UFES is withheld from label supervision during training. PAD-UFES serves as the adversarial target domain during training and therefore is not independent of the optimization process: the domain-adversarial branch of the audited architecture explicitly optimizes the representation to reduce discriminability between ISIC and PAD-UFES.

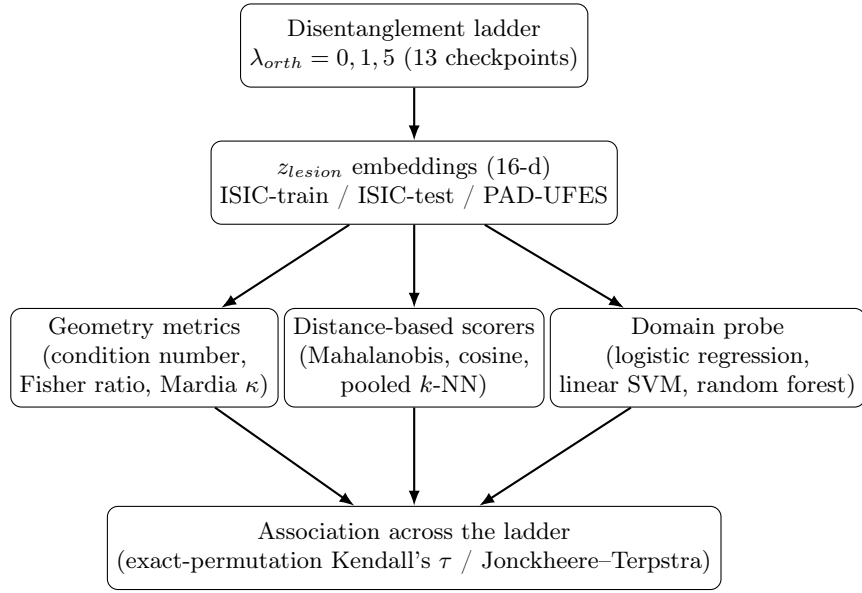


Figure 1: Study design. A disentanglement dose-response ladder (three checkpoint families, $\lambda_{orth} = 0, 1, 5$, sharing architecture and representation) is used to test whether an implicit assumption—that representation geometry reflects reliability-estimation quality—survives a training intervention that reshapes that geometry. Geometry metrics, three distance-based scorers, and a domain-information probe are computed on the identical embeddings for each of 13 checkpoints (Section 2.1).

2.3. The disentanglement dose-response ladder

We use three checkpoint families from a domain-adversarial disentanglement architecture, each producing a 16-dimensional lesion representation (z_{lesion}). The three families share identical architecture, training recipe, and representation dimensionality, differing only in the strength of an orthogonality regularization penalty (λ_{orth}) applied between the lesion and context branches (Table 1).

Table 1: The disentanglement dose-response ladder.

Family	λ_{orth}	Mechanism	Seeds available
runA_gr1	0	domain-adversarial component only	42, 52, 62, 72, 82 (5)
runB_orth1	1	+ orthogonality term	42, 52, 62, 72, 82 (5)
runB	5	+ stronger orthogonality	42, 52, 62 (3)

13 (family, seed) checkpoints total. All checkpoints are referenced by explicit, individually verified file path; none is resolved by automated directory search, to avoid a documented failure mode in which the most-recently-modified file in a run directory is not the best-performing validation epoch.

We additionally report a single conventional (non-disentangled) ResNet-50 checkpoint, **baseline_soft**, differing from the ladder in architecture, representation dimensionality (2048-d vs. 16-d), and training recipe, as a descriptive reference rather than a fourth dose level.

2.4. Representation geometry metrics

The selected metrics were pre-specified to represent three complementary aspects of representation geometry—covariance conditioning, class separation, and multivariate normality—rather than to maximize empirical correlation with downstream performance.

- **Condition number:** the ratio of the largest to smallest eigenvalue of the regularized, shared within-class covariance matrix used to fit the Mahalanobis estimator (Section 2.5), measuring how numerically stable that covariance’s inversion is.
- **Fisher ratio:** the Hotelling–Lawley trace, $\text{tr}(\Sigma^{-1}S_B)$, computed from the same fitted precision matrix, measuring class separation relative to within-class spread. We additionally report a decoupled scalar companion, $\text{tr}(S_B)/\text{tr}(S_W)$, which involves no matrix inverse and is therefore not entangled with condition number’s own estimation noise—the two are reported together because they are not statistically independent measurements when both derive from the same Σ^{-1} .
- **Mardia’s multivariate kurtosis:** quantifies departure from the class-conditional Gaussian shape the Mahalanobis estimator assumes. The classical closed-form asymptotic null does not apply once class means and covariance are estimated in-sample from the same residuals being tested; its

null distribution is calibrated by parametric bootstrap (200 resamples per checkpoint) rather than the textbook formula.

All three metrics are computed on the identical fitted Mahalanobis parameters used for scoring (Section 2.5), not from an independently re-estimated covariance, so that geometry and reliability estimation describe the same fitted object.

2.5. Reliability scorers

Each test embedding is scored by eight structurally distinct rules, applied to identical embeddings and an identical ISIC-train / ISIC-test / PAD-UFES split, for every one of the 13 primary-ladder checkpoints. Five operate only on embedding geometry:

- **Mahalanobis distance:** the minimum, over the 8 ISIC classes, of the squared Mahalanobis distance from the test embedding to each class mean, under a shared covariance fit on the ISIC training set with regularization $\varepsilon = 10^{-5}$.
- **Cosine-to-centroid:** the minimum, over the same 8 class means, of $1 - \cos(z, \mu_c)$ —identical class-centroid structure to the Mahalanobis scorer, with the covariance/precision step removed, isolating that one variable.
- **Pooled k -nearest-neighbor distance:** the Euclidean distance from a test embedding to its k -th nearest neighbor in the full, class-pooled ISIC training set—no class structure, no covariance, the most assumption-free of the five distance-based scorers [?]. $k = 10$ was specified as the primary/headline value in the analysis plan before any k -NN result was computed; $k = 1$ and $k = 50$ are reported as a prespecified robustness grid, not selected after the fact.

Three additional scorers, not distance-based, were evaluated to test whether the failure documented here is specific to distance-based scoring:

- **Energy** [?]: $-\log \sum_k \exp(\text{logit}_k)$, computed from the classifier head’s logits rather than from embedding geometry; a lower log-sum-exp (higher score, in our OOD-is-higher convention) indicates lower classifier confidence.
- **ViM (Virtual-Logit Matching)** [?]: combines a residual-subspace term—the embedding’s norm outside the top principal subspace spanned by the classifier head’s weight matrix—with the same logits Energy uses, calibrated on the training set so the two terms are on a comparable scale.
- **Density (per-class KDE):** a Gaussian-kernel density estimate fit separately on each of the 8 ISIC classes’ training embeddings (bandwidth by Scott’s rule), scored by the negative log-likelihood under the best-fitting class model—replacing Mahalanobis’s single-Gaussian assumption with a non-parametric one, while keeping the same per-class-then-best structure.

Logits for Energy and ViM were reconstructed from each checkpoint’s own classifier head (a linear layer applied directly to the cached z_{lesion} embedding, without a fresh forward pass through the network) and verified to reproduce that checkpoint’s own classification accuracy before use. All eight scorers share one score convention (higher score, more out-of-distribution) and are evaluated by AUROC and FPR-at-95%-TPR on the same ISIC-test-vs-PAD-UFES split. The three non-distance-based scorers were evaluated on the 13 primary-ladder checkpoints only; **baseline_soft** (Section 2.3) uses a different architecture and classifier head and was not included in this extension.

2.6. Domain-information probe

To test whether domain-discriminative information is present in an embedding independent of whether any of the eight scorers above can access it, we train three supervised classifiers—logistic regression, a linear support vector machine, and a random forest, each at library-default hyperparameters with no tuning—to predict domain membership (ISIC-test vs. PAD-UFES) directly from the same z_{lesion} embeddings scored in Section 2.5. None of the three probes has access to the original training objective; each is a fresh classifier fit only for this analysis. Probe AUROC is reported as evidence of what is linearly and nonlinearly recoverable from the representation, not as a claim about the training objective’s internal computation [?].

We report 5-fold stratified cross-validated AUROC on out-of-fold predictions, avoiding the same-data-fit-and-evaluate leakage that would otherwise inflate apparent decodability at this sample size and dimensionality. The ISIC-test/PAD-UFES split is class-imbalanced (approximately 2.2:1); AUROC was selected as the primary probe metric specifically because it is threshold-independent and substantially less sensitive to this imbalance than accuracy-based metrics would be, though it was not supplemented here with an imbalance-corrected metric such as balanced accuracy or precision-recall AUC.

2.7. Statistical testing

The disentanglement ladder has only three ordinal levels with unequal per-level seed counts (5, 5, 3). We test association using Kendall’s τ and, for ordered-group trend testing, the Jonckheere–Terpstra statistic, both evaluated against a full-enumeration exact permutation null—every distinct label arrangement consistent with the true per-rung seed counts—rather than the standard asymptotic approximation, which is not assumed valid at this sample size without checking.

Given the limited number of independent checkpoints ($n = 13$), we emphasize consistency across independent analyses rather than statistical significance from any single test; unadjusted exact p -values are reported without family-wise correction. The analysis plan fixed before any experiment ran specifies a single success criterion for a trend to count as supporting the mechanistic hypothesis: $|\tau| \geq 0.3$ and significance at $\alpha = 0.05$, with the sign stable between the full ladder and the common-seed subset. We state its detectability consequences in Section 2.8.

Kendall’s τ p -values are two-sided throughout. Jonckheere–Terpstra p -values are one-sided upper-tail, the conventional reading of an ordered-alternative trend test; at these group sizes the two-sided Jonckheere–Terpstra p -value is identical to the Kendall’s τ p -value, because with a tie-free response the two statistics are related by the exact identity $\tau_b = (2J - M)/\sqrt{Mn_0}$, where M is the number of cross-rung pairs and n_0 the total number of pairs. The two tests are therefore not independent evidence of the same trend.

2.8. Detectability, power, and interval estimates

A p -value above α states only that the observed statistic was not extreme under the null; it does not say which effect sizes the design could have detected, nor which the data exclude. We therefore report three additional quantities for every association tested, so that each null result carries an explicit bound rather than a qualitative hedge.

First, for each design we compute τ_{crit} , the smallest $|\tau|$ attainable at that design whose two-sided exact permutation p -value reaches 0.05. This is a property of n , the group sizes, and the tie structure alone: no data, however clean, can yield a significant association below it. Second, we compute power by simulation (2×10^4 datasets per point, common random numbers across the effect-size grid) under alternatives parameterized so that the *expected value of the reported statistic* equals the target τ —a bivariate-normal alternative with $\mathbb{E}[\tau] = (2/\pi) \arcsin \rho$ for the continuous–continuous design, and a linear dose-shift alternative $y_i = \theta d_i + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, 1)$, with $\mathbb{E}[\tau_b]$ available in closed form, for the ordered-ladder design. Third, we report a 95% bias-corrected and accelerated (BCa) bootstrap interval on each observed τ , using 2×10^4 resamples drawn *within* rung so that the 5/5/3 design is held fixed — λ_{orth} is a fixed design factor, not a sampled one. Where an observed τ attains the maximum value its design permits, the resampling distribution is one-sided by construction and no bootstrap interval is valid; such cases are reported as such rather than given an interval.

Full results are in Supplementary Sections S8 (design-level critical values, power curves) and S9 (per-test bootstrap intervals).

2.9. Reproducibility

Every checkpoint, seed, and preprocessing parameter used in this study is recorded by explicit value or file path; no result in this paper depends on a directory-resolved or otherwise ambiguously-selected checkpoint. Analysis code and derived results supporting this study are available as described in the Data Availability statement accompanying this submission.

3. Results

3.1. Representation geometry changes with disentanglement strength

Condition number increased monotonically with λ_{orth} : 75.5 ± 7.8 (`runA_gr1`), 559.8 ± 125.9 (`runB_orth1`), 5329.9 ± 2672.2 (`runB`)—a two-order-of-magnitude

change across the ladder (Table 2, Figure 2). This trend was significant by exact-permutation Kendall’s τ ($\tau = 0.84$, $p = 2.8 \times 10^{-5}$, $n = 13$) and sign-stable on the common-seed subset ($\tau = 0.87$, $p = 0.0012$, $n = 9$), and confirmed by Jonckheere–Terpstra trend testing ($J = 55.0$ against a null mean of 27.5, exact $p = 1.4 \times 10^{-5}$).

The remaining four geometry metrics showed no significant trend: Fisher ratio (Hotelling–Lawley), $\tau = 0.17$, $p = 0.52$; Fisher ratio (decoupled scalar), $\tau = 0.08$, $p = 0.80$; Mardia’s kurtosis (b), $\tau = -0.14$, $p = 0.61$; Mardia’s kurtosis (z), $\tau = -0.17$, $p = 0.52$ (Table 2).

Increasing λ_{orth} is associated with a large, statistically significant change in condition number; the other four geometry metrics tested show no significant association with λ_{orth} .

Table 2: Representation geometry across the disentanglement ladder (mean $\pm$ SD).

Metric	runA_grl ($\lambda_{orth} = 0$)	runB_orth1 ($\lambda_{orth} = 1$)	runB ($\lambda_{orth} = 5$)
Condition number	75.5 ± 7.8	559.8 ± 125.9	5329.9 ± 2672.2
Fisher ratio (Hotelling–Lawley, $\times 10^5$)	5.41 ± 0.81	5.71 ± 0.43	5.63 ± 0.50
Fisher ratio (decoupled scalar)	4.69 ± 0.65	4.94 ± 0.37	4.95 ± 0.38
Mardia’s kurtosis (b)	739.9 ± 86.5	788.0 ± 240.9	679.9 ± 63.7
Mardia’s kurtosis (z)	341.1 ± 99.3	397.3 ± 257.6	270.7 ± 53.4
n seeds	5	5	3

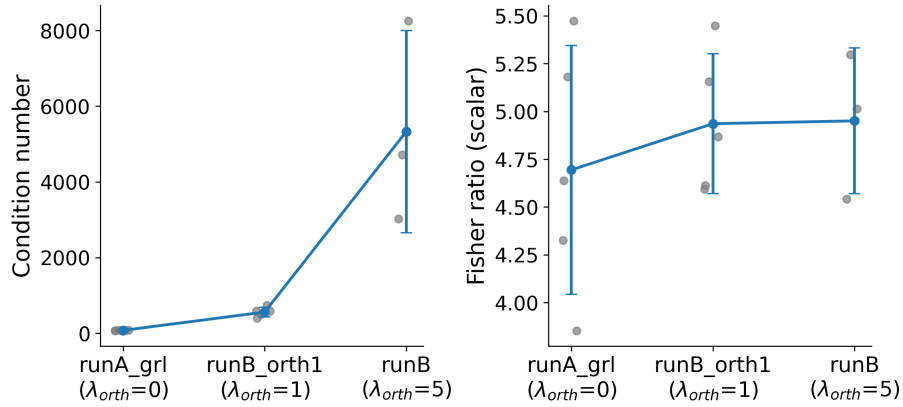


Figure 2: Representation geometry across the disentanglement ladder. Condition number (left) and the decoupled Fisher-ratio scalar $\text{tr}(S_B)/\text{tr}(S_W)$ (right) for each of 13 checkpoints (gray points; mean $\pm$ SD in blue), across three levels of orthogonality strength. Condition number increases by approximately two orders of magnitude with λ_{orth} (Kendall’s $\tau = 0.84$, exact $p = 2.8 \times 10^{-5}$; Table 2); the Fisher-ratio scalar shows no significant trend. The remaining three geometry metrics (Fisher ratio, Hotelling–Lawley form; Mardia’s kurtosis b and z) are in Supplementary Figure S1.

3.2. Representation geometry shows no measurable association with Mahalanobis AUROC

Mahalanobis AUROC (ISIC-test vs. PAD-UFES) was 0.401 ± 0.028 (**runA_grl**), 0.408 ± 0.022 (**runB_orth1**), and 0.393 ± 0.025 (**runB**)—below the chance value of 0.5 at every rung (Table 3).

None of the five geometry metrics, including condition number, showed a significant association with Mahalanobis AUROC: condition number, $\tau = -0.13$, $p = 0.59$; Fisher ratio (HL), $\tau = -0.28$, $p = 0.20$; Fisher ratio (scalar), $\tau = -0.15$, $p = 0.51$; Mardia’s kurtosis (b), $\tau = -0.13$, $p = 0.59$; Mardia’s kurtosis (z), $\tau = -0.13$, $p = 0.59$ (Figure 3). AUROC itself showed no significant trend across the ladder (Jonckheere–Terpstra, $J = 25.0$ against a null mean of 27.5, exact $p = 0.65$).

An approximately two-orders-of-magnitude change in condition number (Section 3.1) is not accompanied by a significant change in Mahalanobis AUROC; no geometry metric tested shows a significant association with AUROC.

Table 3: Mahalanobis-distance AUROC (ISIC-test vs. PAD-UFES), by rung (mean $\pm$ SD across seeds).

Rung	λ_{orth}	n seeds	AUROC	FPR@95%TPR
runA_grl	0	5	0.401 ± 0.028	0.985 ± 0.009
runB_orth1	1	5	0.408 ± 0.022	0.980 ± 0.012
runB	5	3	0.393 ± 0.025	0.985 ± 0.003

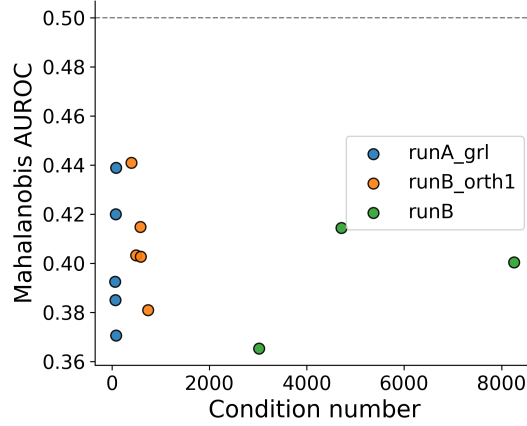


Figure 3: Representation geometry does not predict Mahalanobis AUROC. Condition number vs. Mahalanobis-distance AUROC (ISIC-test vs. PAD-UFES), one point per checkpoint, colored by rung. No association is apparent (Kendall’s $\tau = -0.13$, $p = 0.59$; Table 3); the remaining four geometry metrics show the same pattern (Supplementary Figure S2).

3.3. Distance-based scores are systematically lower for the target domain

Median Mahalanobis squared distance was lower for PAD-UFES than for ISIC-test at every one of the 13 checkpoints. Pooled across seeds within each rung: **runA_gr1**, ISIC-test median 13.50 vs. PAD-UFES median 8.57 ($n_{id} = 25335$, $n_{ood} = 11490$); **runB_orth1**, 14.10 vs. 8.99 ($n_{id} = 25335$, $n_{ood} = 11490$); **runB**, 13.20 vs. 8.08 ($n_{id} = 15201$, $n_{ood} = 6894$). Median distance was lower for the target domain than the source domain at all 13 checkpoints.

Feature-norm distributions differed modestly between domains. Pooled median $\|z\|$: **runA_gr1**, ISIC-test 6.31 vs. PAD-UFES 6.21; **runB_orth1**, 6.38 vs. 6.28; **runB**, 6.28 vs. 5.97. At the level of pooled means, the direction was not consistent across rungs (**runA_gr1** mean: 6.72 ISIC-test vs. 6.83 PAD-UFES). The direction of this difference was not consistent at the level of means.

Nearest-centroid classification assigned PAD-UFES samples to the majority ISIC class (Nevus) at 61.6% (**runA_gr1**), 61.9% (**runB_orth1**), and 63.9% (**runB**), compared to 53.0%, 53.1%, and 53.4% respectively for ISIC-test samples classified against their own nearest centroid. This was 8.7–10.5 percentage points higher than the ISIC-test baseline rate (ratio 1.16–1.20).

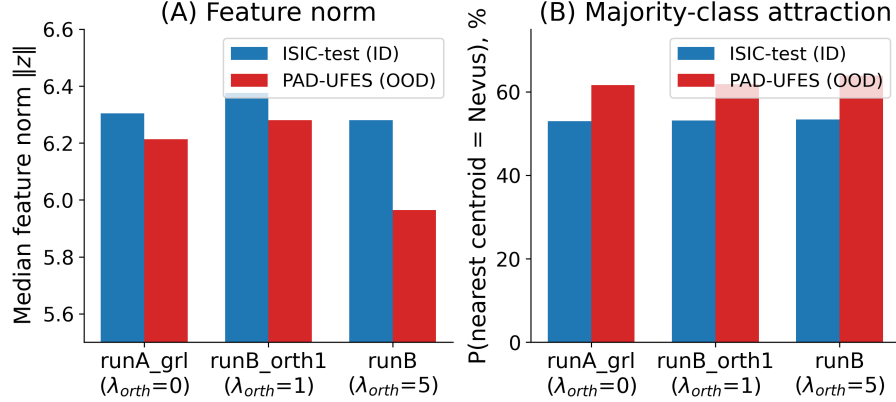


Figure 4: Neither feature-norm difference nor majority-class attraction fully accounts for the raw ID/OOD distance gap. (A) Pooled median feature norm, ISIC-test vs. PAD-UFES, per rung. (B) Nearest-centroid assignment of ISIC-test and PAD-UFES samples to the majority ISIC class (Nevus), per rung. The norm difference is modest and not consistent in direction at the level of means; PAD-UFES’s excess attraction to the majority class (8.7–10.5 percentage points above the ISIC-test baseline) is real but does not, on its own, account for the full distance separation in Figure 5.

3.4. The failure generalizes across eight scorer families

Pooled across all 13 checkpoints, AUROC was 0.402 ± 0.024 (Mahalanobis), 0.418 ± 0.022 (cosine-to-centroid), 0.424 ± 0.023 (k -NN, $k = 1$), 0.414 ± 0.024 (k -NN, $k = 10$), and 0.411 ± 0.023 (k -NN, $k = 50$) (Table 4, Figure 5). All five values fall within a 0.40–0.42 range and below the chance value of 0.5. Despite

relying on substantially different scoring formulations—parametric with covariance, parametric without covariance, and non-parametric—all five distance-based scorer families produced highly similar AUROC values.

To test whether this failure is specific to distance-based scoring, we additionally evaluated three scorers that are not distance-based: Energy (a logit-based confidence score), ViM (Virtual-Logit Matching, a residual-subspace scorer that combines geometry and logits), and a per-class kernel density estimator. Pooled across the same 13 checkpoints, AUROC was 0.447 ± 0.047 (Energy), 0.387 ± 0.022 (ViM), and 0.417 ± 0.025 (density estimator)—all three within the same 0.39–0.45 range as the five distance-based scorers, and below or near the chance value of 0.5.

None of the five distance-based scorer variants showed a significant association with λ_{orth} : Mahalanobis, $\tau = -0.08$, $p = 0.80$; cosine, $\tau = 0.32$, $p = 0.19$; k -NN ($k = 1$), $\tau = -0.05$, $p = 0.90$; k -NN ($k = 10$), $\tau = 0.02$, $p = 1.00$; k -NN ($k = 50$), $\tau = 0.08$, $p = 0.80$. Jonckheere–Terpstra trend testing gave the same result for all five (exact p between 0.10 and 0.65). Among the three additional scorers, ViM ($\tau = 0.08$, $p = 0.80$) and the density estimator ($\tau = 0.02$, $p = 1.00$) likewise showed no significant association. Energy exhibited a modest monotonic trend with increasing disentanglement (Kendall’s $\tau = 0.44$, exact $p = 0.065$; Jonckheere–Terpstra $p = 0.032$), whereas the remaining seven scorers showed no consistent trend (full statistics in Supplementary Table S7).

Eight structurally distinct scorers—five distance-based (parametric with and without covariance, and non-parametric k -NN at three values of k) and three that are not distance-based (a logit-based confidence score, a residual-subspace scorer, and a non-parametric density estimator)—report AUROC in essentially the same 0.39–0.45 range, all at or below chance, with one exception (Energy’s Jonckheere–Terpstra trend) discussed in Section 4.

Table 4: AUROC (ISIC-test vs. PAD-UFES) for eight scorers, by rung and pooled (mean $\pm$ SD across seeds).

Scorer	runA_gr1	runB_orth1	runB	Pooled ($n = 13$)
<i>Distance-based</i>				
Mahalanobis	0.401 ± 0.028	0.408 ± 0.022	0.393 ± 0.025	0.402 ± 0.024
Cosine-to-centroid	0.411 ± 0.020	0.413 ± 0.015	0.439 ± 0.027	0.418 ± 0.022
k -NN ($k = 1$)	0.424 ± 0.028	0.428 ± 0.020	0.416 ± 0.028	0.424 ± 0.023
k -NN ($k = 10$)	0.413 ± 0.029	0.417 ± 0.020	0.408 ± 0.029	0.414 ± 0.024
k -NN ($k = 50$)	0.411 ± 0.027	0.414 ± 0.020	0.408 ± 0.030	0.411 ± 0.023
<i>Non-distance-based</i>				
Energy	0.429 ± 0.044	0.429 ± 0.022	0.506 ± 0.035	0.447 ± 0.047
ViM	0.381 ± 0.007	0.392 ± 0.036	0.386 ± 0.014	0.387 ± 0.022
Density (KDE)	0.416 ± 0.029	0.420 ± 0.022	0.412 ± 0.031	0.417 ± 0.025

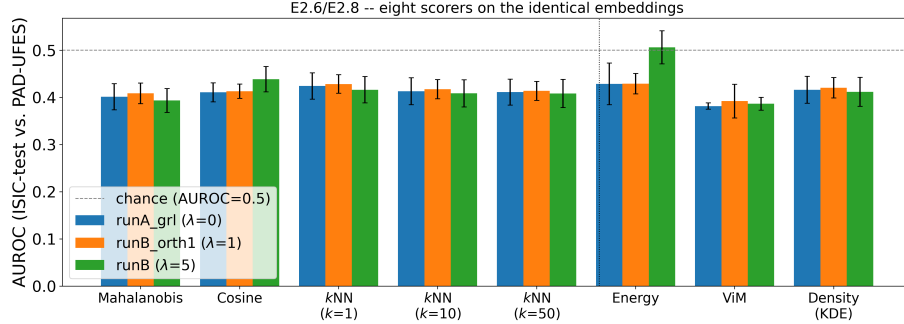


Figure 5: Eight scorers fail alike, with one isolated exception. AUROC (ISIC-test vs. PAD-UFES) for five distance-based scorers (Mahalanobis distance, cosine-to-centroid similarity, pooled k -nearest-neighbor distance at $k = 1, 10, 50$) and three non-distance-based scorers (Energy, ViM, per-class KDE density), computed on the identical embeddings, grouped by rung (mean $\pm$ SD across seeds). Dashed line: chance (AUROC = 0.5); dotted line separates the two scorer families. Seven of eight scorers fall within a 0.39–0.42 range with no significant trend; Energy alone rises with λ_{orth} (Jonckheere–Terpstra exact $p = 0.032$), discussed in Section 4 as an isolated, hypothesis-generating observation (Table 4).

3.5. Domain information remains decodable despite distance-based inaccessibility

On the identical embeddings scored in Section 3.4, three supervised probes recovered domain membership (ISIC-test vs. PAD-UFES) at: **runA_gr1**, 0.719 (logistic regression), 0.717 (linear SVM), 0.807 (random forest); **runB_orth1**, 0.741, 0.740, 0.815; **runB**, 0.750, 0.751, 0.800 (Table 5, Figure 6). AUROC exceeded 0.70 for all three probe families at every rung.

None of the three probes showed a significant association with λ_{orth} : logistic regression, $\tau = 0.26$, $p = 0.30$; linear SVM, $\tau = 0.26$, $p = 0.30$; random forest, $\tau = -0.11$, $p = 0.70$.

Three independent classifiers, applied to the same embeddings scored by the eight non-probing scorers in Section 3.4, recover domain membership at 0.72–0.81 AUROC at every λ_{orth} level tested; probe AUROC shows no significant association with λ_{orth} .

Table 5: Non-probing scorer AUROC (pooled across Table 4) vs. domain-probe AUROC, by rung. **baseline_soft** is a descriptive reference, not a fourth ladder point (Section 2.3).

Rung	Role	Scorers (pooled)	Probe (LR)	Probe (SVM)	Probe (RF)
runA_gr1	primary ladder	0.411	0.719	0.717	0.807
runB_orth1	primary ladder	0.415	0.741	0.740	0.815
runB	primary ladder	0.421	0.750	0.751	0.800
baseline_soft	categorical reference ($n = 1$)	0.828	0.998	0.997	0.977

3.6. Descriptive reference: a non-disentangled baseline

For a single ResNet-50 checkpoint (**baseline_soft**; one seed, architecture and representation dimensionality differing from the ladder, Section 2.3), pooled

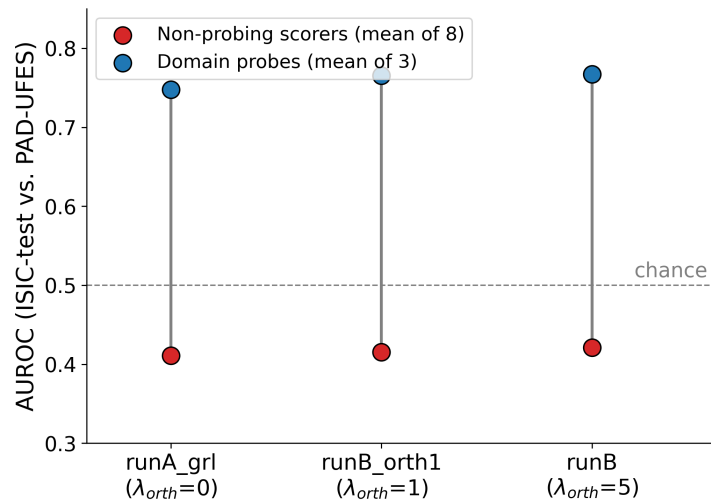


Figure 6: Domain information is decodable but not accessible to non-probing scorers. For each rung, mean AUROC across the eight scorers in Figure 5 (five distance-based, three non-distance-based) is connected to mean AUROC across three independent domain probes—logistic regression, linear SVM, random forest (Section 2.6)—computed on the identical embeddings. Non-probing scorers cluster at or below chance (≈ 0.41); probes recover domain membership at 0.72–0.81 AUROC at every λ_{orth} level. **baseline_soft** (Section 3.6) is not plotted on this axis, as it is a descriptive reference confounded by architecture and dimensionality (Section 2.3), not a controlled comparison point.

AUROC across the five distance-based scorers was 0.828 (the three non-distance-based scorers were not evaluated on this architecture, Section 3.4), and probe AUROC was 0.998 (logistic regression), 0.997 (linear SVM), and 0.977 (random forest) (Table 5).

4. Discussion

4.1. Finding

Increasing disentanglement strength substantially changed representation geometry without improving distance-based reliability estimation, while the domain information such estimation would require remained recoverable from the same representations by a non-distance-based reader.

4.2. Interpretation

This study audited a simple assumption: changing representation geometry should change the behavior of distance-based reliability estimators. The first part of this assumption held—condition number moved by two orders of magnitude across the disentanglement ladder (Section 3.1). The second did not: no geometry metric tracked Mahalanobis AUROC, and AUROC itself showed no trend across the ladder (Section 3.2).

A null result admits two straightforward explanations: that Mahalanobis distance specifically is a poor instrument, or that disentanglement training erased the domain-discriminative information a reliability estimator would need. Neither holds. Cosine-to-centroid similarity discards the covariance structure Mahalanobis distance depends on; pooled k -nearest-neighbor distance discards the notion of a class centroid altogether; Energy [?] discards representation geometry altogether in favor of the classifier’s own logits; ViM [?] combines a residual-subspace geometric term with those same logits; a per-class kernel density estimator replaces Mahalanobis’s single-Gaussian assumption with a non-parametric one. Despite these substantially different scoring principles, seven of the eight scorers tested converged on the same 0.39–0.42 AUROC (Section 3.4)—a failure shared across formulas this different reflects what they share, not a defect specific to any one. Nor was the relevant information absent: three probes with no access to the training objective recovered domain membership at 0.72–0.81 AUROC at every disentanglement level tested (Section 3.5), well above what these scoring rules reported on the identical embeddings. The information was not gone. It was present, recoverable, and specifically unreachable by seven of the eight scoring rules tested.

The eighth scorer, Energy, is an exception we report plainly rather than explain away. Among the eight evaluated scorers, Energy exhibited a modest monotonic trend with increasing disentanglement (Jonckheere–Terpstra exact $p = 0.032$, one-sided; its two-sided counterpart coincides with the Kendall $p = 0.065$ by the τ_b – J identity of Section 2.7, so the two tests are one piece of evidence read two ways, not two), whereas the remaining seven scorers showed

no consistent trend (Section 3.4). This isolated observation should be interpreted cautiously: it is one result out of eight scorers tested without correction for multiple comparisons, it was not part of the analysis plan fixed before the experiments ran, its Kendall’s τ (0.44, exact $p = 0.065$) does not clear the $|\tau| \geq 0.3$ -and-significant threshold that plan specifies (Section 2.7), and the effect size is modest. We treat it as a hypothesis-generating observation rather than as evidence against the main conclusion, which we accordingly scope to seven of the eight scorers tested rather than all eight. That Energy alone showed this trend suggests that accessibility may depend not only on representation geometry but also on the inductive biases of individual scoring functions—whether this reflects a genuine property of energy-based scoring or a dataset-specific effect is an open question, as is whether it persists under different pretraining paradigms or model architectures.

Taken together, these findings indicate that **information can remain decodable while becoming largely inaccessible to non-probing reliability estimators**. This is not a claim that any one of these methods is poorly designed, and not a claim that disentanglement training destroys information. It is a claim about a gap between two properties of a representation—whether information is present, and whether it is organized in a form a specific family of estimators can use—that the ladder design in this study makes it possible to separate for the first time in this setting. This claim is scoped to the training regime studied here—domain-adversarial disentanglement evaluated against its own adversarial target domain (Section 4.4)—not to representation learning or domain shift in general.

Section 3.3’s supporting evidence (a modest, direction-inconsistent norm difference; a majority-class attraction of 8.7–10.5 percentage points above baseline) indicates that the raw distance gap between domains is real but is not fully accounted for by either mechanism individually; neither is proposed here as a complete account of *why* the gap exists. Understanding the origin of this systematic below-chance behavior remains future work.

4.3. Relation to literature

Mahalanobis-distance OOD detection [?] and its non-parametric successors, including pooled k -nearest-neighbor scoring [?], are typically evaluated by benchmarking AUROC across datasets and architectures, not by auditing whether the geometric assumption connecting representation and reliability estimate survives a specific training intervention. Energy-based scoring [?] and Virtual-Logit Matching [?] were developed specifically to move beyond a purely geometric notion of reliability by incorporating the classifier’s own logits, making them a natural extension of the audit rather than a different question. This study does not propose a competitor to any of these methods; all five distance-based and three non-distance-based scorers are used here as instruments for the audit, alongside a cosine-similarity variant and a kernel density estimator that isolate specific structural assumptions from one another. The finding that seven of the eight degrade identically under this training regime is a statement about a premise most of these methods share, not a comparative ranking of them.

Domain-adversarial and orthogonality-based disentanglement training [?] is one of several strategies proposed to reduce reliance on shortcut features in deep classifiers [?]. That literature evaluates disentanglement primarily by its effect on classification robustness and shortcut reliance; this study evaluates a different, downstream consequence—what disentanglement does to a reliability estimator built on top of the resulting representation—and finds that the two do not move together. This does not bear on whether disentanglement succeeds at its stated goal, only on whether a distance-based reliability estimator layered on top of it continues to function as it did before the intervention.

The distinction this study draws between the presence of information in a representation and a specific estimator’s ability to use it is closely related to, and constrained by, the probing-classifier literature’s own caution that a supervised probe’s success does not establish that a model itself uses the decoded information in the way the probe accesses it [?]. This study makes a narrower claim than that literature warns against overreaching toward: the probes here are not offered as evidence about what the disentanglement objective computes internally, only as an independent measurement of what remains extractable from its output representation, without characterizing the mechanism by which that representation was produced.

More broadly, these findings suggest that evaluating learned representations only through downstream predictive performance may overlook an equally important question: whether the representation is organized in a form that downstream reliability estimators can actually exploit. Geometry, decodability, and distance accessibility therefore appear to be related but distinct properties, and improvement in one should not automatically be interpreted as improvement in the others.

For practitioners combining domain-adversarial disentanglement training with a distance-based reliability gate in a deployed pipeline, this audit implies a specific caution: such a gate may not reliably flag inputs from the domain the disentanglement objective was trained against, even where it performs as expected against unrelated distributions. Where feasible, reliability estimation in such pipelines should be validated directly against the domain the training procedure targets, rather than assumed to transfer from general model qualification.

4.4. Limitations

PAD-UFES is not an arbitrary target domain in this design: it is the domain the disentanglement architecture’s adversarial branch is trained against. Since PAD-UFES also serves as the adversarial domain during training, our conclusions should be interpreted as characterizing representations learned under this training regime rather than all domain shifts.

The comparison against a conventional, non-disentangled baseline (Section 3.6) differs from the disentanglement ladder in architecture, representation dimensionality, and training recipe simultaneously, and is drawn from a single checkpoint. It is reported descriptively and does not support any causal attribution

of the ladder’s findings to disentanglement specifically, as opposed to architecture or dimensionality; a matched, non-adversarial checkpoint at the same architecture and 16-dimensional representation would be needed to make that attribution, and no such checkpoint exists in the inventory available for this study. Attributing the central finding specifically to disentanglement, rather than to architecture or dimensionality, remains an important direction.

The ladder itself spans three ordinal treatment levels with unequal per-level seed counts ($n = 13$ total, $n = 9$ on the common-seed subset), and we state the resulting limits on detection quantitatively rather than as a qualitative caveat (Section 2.8; Supplementary Section S8). At $n = 13$ the smallest $|\tau|$ that can reach $p \leq 0.05$ under the exact permutation null is 0.436 for the continuous–continuous design used in Section 3.2 and 0.473 for the 5/5/3 ordered ladder used in Sections 3.4–3.5 (0.609 on the $n = 9$ common-seed subset). Two consequences follow, and we state both explicitly. The pre-specified success criterion this study applies— $|\tau| \geq 0.3$ and significant—lies below all three of those thresholds and was therefore unsatisfiable before any data were collected: an association of exactly the pre-specified size could not have been declared significant at this design. And the power to detect an expected $\tau = 0.3$ is 0.27, 0.23, and 0.14 for the three designs respectively; 80% power is first reached at $\tau = 0.54$, 0.57, and 0.69. The null results in this study are consequently informative about large associations and close to uninformative about moderate ones.

The bootstrap intervals in Supplementary Section S9 say the same thing from the data side. Of the twenty null associations for which a bootstrap interval is valid, nineteen have a 95% interval wide enough to contain $|\tau| = 0.3$; the sole exception is ViM ($\tau = 0.076$, 95% CI $[-0.268, 0.270]$), which is the only null result here that excludes an association of the pre-specified magnitude. For the remaining nineteen the honest statement is that this study does not distinguish “no association” from “a moderate association this design cannot resolve.” We report this because it bounds the central claim: the dissociation documented in Sections 3.2–3.5 rests on how consistently the point estimates fall near zero across the twenty associations tested outside the condition-number effect—four further geometry metrics against λ_{orth} , five against Mahalanobis AUROC, eight scorers, and three probes—and not on any individual interval being tight. Eighteen of those twenty have $|\tau| < 0.3$; the two that do not (Energy, $\tau = 0.443$; cosine-to-centroid, $\tau = 0.321$) are reported as such in Section 3.4 and are not counted as null. These twenty tests are not mutually independent—the geometry metrics derive from a common fitted precision matrix and the three probes share one feature set—so their agreement is weaker evidence than twenty independent replications would be. A single one of these tests, read alone, would not support the claim.

Exact-permutation testing was used throughout in place of asymptotic approximations not verified to hold at this scale, and p -values are reported without family-wise correction across the multiple metrics and scorers tested, on the stated basis that conclusions rest on consistency of effect across independent analyses rather than on any single threshold crossing; a reader applying a stricter correction should note that the one large, load-bearing effect in this

study (condition number’s association with λ_{orth} , $\tau = 0.840$ —which is the maximum value the 5/5/3 design permits, i.e. a perfectly monotone ladder) survives any standard correction, while every null result was already reported as non-significant before any correction was considered.

This study does not include a replication on a domain independent of the ISIC/PAD-UFES pair, nor a check of generalization beyond the single architecture audited here. Both were considered and are not included: a third-domain replication would require a dataset without documented image overlap with the training data, which was not available within the scope of this study, and a cross-architecture check was not pursued for the same reason. Neither omission is filled by the baseline reference described above, which addresses architecture and dimensionality but not domain independence. The bootstrap check proposed elsewhere to quantify shared estimation noise between condition number and Fisher ratio, both derived from the same fitted precision matrix, was not performed; condition number, not Fisher ratio, is the metric driving the geometric finding in Section 3.1, which bounds but does not eliminate this as a source of uncertainty in the broader geometry measurement. Checkpoint selection was verified by explicit file path rather than by directory search, but was not independently cross-checked against logged validation curves beyond the single file available per run. Mardia’s kurtosis null distribution was calibrated with 200 bootstrap resamples per checkpoint (Section 2.4); the Mahalanobis regularization constant ($\epsilon = 10^{-5}$) and the 16-dimensional representation itself were inherited from the audited classifier rather than varied as part of this study. “Reliability estimation” throughout this study refers specifically to out-of-distribution detection AUROC; broader constructs such as calibration or selective prediction are not addressed.

This distinction is particularly relevant when domain-adversarial representations are combined with distance-based or logit-based uncertainty estimation in safety-critical medical AI systems. The same information can remain decodable while becoming largely inaccessible to non-probing reliability estimators.

5. Conclusion

This study audited whether disentanglement training that reshapes representation geometry preserves the assumption distance-based reliability estimation depends on. It largely does not: representation geometry changed substantially, and reliability estimation largely did not track it—a failure shared by seven of eight structurally distinct scoring rules tested, including three that are not distance-based. The eighth, an energy-based scorer, showed an isolated trend that we treat as hypothesis-generating rather than as evidence against this pattern (Section 4). The information these scorers would have needed remained recoverable from the same representations by a supervised probe, indicating that the two properties—whether information is present in a representation and whether it is organized in a form a given estimator can use—can diverge under this training regime. Evaluating learned representations for downstream reliability estimation may therefore require assessing both properties directly, rather

than assuming one implies the other. Whether this gap persists under different pretraining paradigms or model architectures remains an open question for future work.

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Declarations

Ethical approval

This study used publicly available de-identified datasets; therefore, ethical approval was not required.

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Competing interests

The author declares no competing interests.

CRedit authorship contribution statement

D.-V. Tran contributed to conceptualization, methodology, software, validation, formal analysis, investigation, resources, data curation, visualization, writing—original draft, and writing—review and editing.

Data availability

Both datasets used in this study are publicly available. The ISIC 2018 dataset is accessible through the ISIC Archive (<https://challenge.isic-archive.com/data/>). The PAD-UFES-20 dataset is available at <https://data.mendeley.com/datasets/zr7vgbcyr2>. Code will be made publicly available upon acceptance.